Endocrine Organs Lab

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Block: Endo/Uro /Repro

Thread: Anatomy/Histology

Conflict of Interest: Dr Spudich has no conflicts of interest to disclose.

Textbook Links

Go to Chapter 21

1. Pawlina W. Histology : A Text and Atlas with Correlated Cell and Molecular Biology. 9th ed. Wolters Kluwer; 2024.

Learning Objectives

By exam time, you should be able to do the following:

1. Discuss the structural and functional anatomy of select organs in the endocrine system, including the hormones secreted by each and how they are moved into the blood supply.
2. Identify the cells and tissues of select endocrine organs by histology.
3. Compare and contrast the embryology of select endocrine organs and the impact of embryology on the adult function and structure at the gross and microscopic levels.
4. Predict how different pathologies of select endocrine organs might alter the histological or physiologic features of the component cells/tissues.

Study Questions

1. Compare and contrast the embryonic origins of the pituitary lobes with respect to their histological appearance.

2. One of your patients has a newly discovered mutation that drives the expression of 17-α-hydroxylase expression in the Zona glomerulus. What might be some metabolic consequences?

Biolucida Slide Box Hyperlink

Folder: Microscopic Anatomy Student Slides; JeffMD Lab #12: Endocrine

Pituitary-1, Feline, H&E	Parathyroid Gland-1, Human, H&E
Pituitary-2, Human, H&E	Parathyroid Gland-2, Human, H&E
Thyroid Gland-1, Human, H&E	Adrenal Gland-1, UD, H&E
Thyroid Gland-2, Monkey, H&E, 1.5µm	Adrenal Gland-2, Monkey, H&E
Parathyroid/Thyroid Gland, UD, H&E	

A. Overview of Endocrine Organs and System

1. The endocrine system is a group of organs needed to maintain homeostasis through a communication system that uses hormones.
   a. These hormones travel through the interstitial spaces and blood to reach their target organs.
   b. Slow profound changes are manifest.

2. The organs of the endocrine system are epithelioid in nature. The cells are embedded in a connective tissues matrix and contents are released directly into associated blood vessels, no exocrine ducts.
   a. This means most endocrine organs are well vascularized.

3. Embryologic origins of the glands are varied. Most are from the ectoderm or endoderm germ layers.
   a. Only the adrenal cortex and gonads have a mesodermal origin.

4. In addition to discrete organs many systems contain individual cells that serve a local endocrine function.
   a. The Diffuse Neuroendocrine System (DNES) works in paracrine and/or autocrine fashion within these larger systems.

B. Hormones

1. Hormones or hormonally active substances can be many types of compounds.
   a. **Peptides**: small proteins, peptides, or polypeptides. This is the largest group of hormonal compounds. Examples include insulin, luteinizing hormone, growth hormone, ACTH.
      i. Most peptide-based hormones are released directly into the blood, but some are bound to a carrier molecule as they move through the blood stream.

   b. **Steroids**: These are cholesterol-derived signaling molecules. These are made, primarily by the adrenal cortex and the gonadal organs.
      i. Steroids utilize carrier proteins to move through the plasma. They protect the hormone and must be released from the carrier protein for activation.

   c. **Amino Acid analogs**: These include the thyroid hormones and the catecholamines.
      i. Tyrosine is the base for the thyroid hormones and iodine is added in the thyroid gland. Thyroid hormones use carrier proteins.
      ii. Catecholamines are tyrosine and phenylalanine derivatives. Catecholamines dissolve readily into the blood

   d. **Arachidonic acid analogs**: These lipid-based signaling molecules are made from arachidonic acid and secreted by many cells as local mediators. They include the leukotrienes, prostaglandins and prostacyclins.

C. Hormone Receptors

1. Hormones can interact with targets via multiple receptor mediated mechanisms.
2. Cell-surface receptors are used when the hormone cannot move through the plasma membrane of the target cells (i.e., catecholamines). They will bind to a receptor expressed on the surface of the cell and activate a series of second messengers within the cell.

a. G-protein coupled receptors and increases in cAMP or cGMP.

b. Kinase cascades

c. Ion channels

d. IP3 and Calcium waves

3. Intracellular receptors are positioned in the cytosol OR in the nucleus. Hormones that can easily pass into the cell use these. Cytoplasmic receptors are often translocated into the nucleus when bound by their hormone.

   a. In the nucleus they bind transcriptional machinery and affect cell function in slower, more profound ways.

   b. Some hormones using intracellular receptors can impact function more quickly by affecting second messenger systems in the cytosol.

D. Signaling Paradigms

1. These secreted molecules can act in three different ways to exert control over distances or very locally.

   a. **Endocrine** control: molecules are distributed through the blood stream to effect large changes.

   b. **Paracrine** control: secreted hormones travel through interstitial tissue or local vascular networks to drive change in nearly tissues/cells.

   c. **Autocrine** control: When the secreting cell expresses receptors for its own hormones and is controlled by them via feedback mechanisms.

2. Feedback is when the response to a stimulus (hormone) influences the cell that produced the stimulus (endocrine cell)

   a. Negative feedback – when response diminishes the original signal. Most endocrine feedback loops are negative feedback loops.

   b. Positive feedback – when stimulus enhances the original signal.

E. Pituitary (Hypophysis)

1. Attached and functionally linked to the hypothalamus of the brain.
   a. Hangs off the hypothalamic region vie the pituitary stalk or infundibulum.

2. Together they control almost all the other endocrine organs and metabolic pathways. Hence, they are called the *Master Organs* of the endocrine system.
3. It has two lobes: anterior and posterior
   a. Anterior is derived from an out-pocketing of the oropharyngeal ectoderm (Rathke's pouch). It is glandular epithelium. It is divided into three distinct regions: Pars distalis, Pars intermedius, and Pars tuberalis.
   b. The posterior lobe is from neuroectoderm that grows out of the floor of the third ventricle in the brain, it can be subdivided into the Pars nervosa and the infundibulum.

4. A portion of the vascular network through the pituitary gland is classified as a portal system.
   a. Fenestrated capillaries in the median eminence of hypothalamus and the infundibulum collect neuroendocrine secretions from the brain tissue.
   b. Hypophyseal portal veins move the secretions to the pituitary gland (mostly pars distalis) where the secretions move into a 2nd capillary network to act on pituitary cells.
   c. Pituitary cells will react and secrete their own hormones into the microvasculature which will be collected by the hypophyseal veins and moved to the systemic venous system.

   - In **Pituitary-1**, identify the pars distalis, pars intermedia, pars tuberalis and pars nervosa. Between the pars intermedia and pars nervosa, identify the remnants of the lumen of Rathke's pouch. In **Pituitary-2**, identify the areas comprising the pars distalis and pars nervosa.

F. Anterior Pituitary Gland (Adenohypophysis)

1. The anterior lobe is a classic epithelial glandular tissue. Cells are in cords or nests separated by loose stromal tissues, through which run the fenestrated capillaries.
2. They secrete in response to signals coming through the portal vessels from the hypothalamus.
3. Tropic hormones are secreted to regulate the activity of other endocrine organs:
   a. Adrenocorticotropic Hormone (ACTH)
   b. Thyroid Stimulating Hormone (TSH)
   c. Follicle Stimulating Hormone (FSH)
   d. Luteinizing Hormone (LH)

4. Non-tropic hormones are secreted by the anterior lobe and act on NON-endocrine tissues.
   a. Growth Hormone (GH)
   b. Prolactin (PRL)

5. ***PARS DISTALIS*** is the largest portion of the anterior lobe of the pituitary. The cells can be classified, histologically, by size and staining properties. These characteristics DO NOT predict the hormones secreted by each.
   a. Basophils make up 10% of the cellular population. They stain dark due to basophilic granules in their cytoplasm.
   b. Acidophils (40%) of cells. They possess acidophilic granules and so their cytoplasm is more red.
   c. Chromophobes (50%) have clear cytoplasm.

6. True classification based on hormones synthesized require the tissues of the pars distalis to be analyzed by immunohistochemistry.
   a. Somatotropes secrete Growth Hormone (GH) and are usually acidophils. Secretion from somatotropes is regulated by two substances from the hypothalamus; Growth Hormone Releasing Hormone (stimulated release of GH) and Somatostatin (inhibits GH release). GH is also regulated by ghrelin from the gut.
   b. Lactotrophs secrete PRL and are acidophils before secretion and chromophobes after release. Lactotropes undergo hypertrophy and

hyperplasia during pregnancy and lactation, which causes the overall size of the pituitary gland to increase in multiparous patients.

c. Corticotropes secrete proopiomelanocortin (POMC), a precursor to ACTH. They are basophils with centrally placed round nuclei. They would also stain positively with PAS reagent due to the biochemistry of POMC. Release of ACTH from these cells is regulated by the hypothalamic hormone, Corticotropin Releasing Hormone (CRH)

i. POMC is enzymatically processed to several signaling molecules: ACTH, beta-lipotropic hormone, melanocyte stimulating hormone, beta-endorphin, enkephalin.

d. Gonadotropes secrete LH and FSH and stain as basophils and are scattered throughout the pars distalis. Gonadotropes are regulated by the hypothalamus via gonadotropin releasing hormone (GnRH).

e. Thyrotropes secrete thyroid-stimulating hormone (TSH) and stain as basophils. Regulation of TSH release is through the hypothalamic hormone Thyrotropin-releasing hormone (TRH). This hormone can also stimulate secretion of PRL. TSH secretion can be inhibited by somatostatin.

7. Follicular stellate cells are support cells found in the anterior lobe of the pituitary gland. They have numerous cytoplasmic processes that envelope the hormone secreting cells. They communicate vis gap junctions and may work to cluster hormone secreting cells into local microenvironments.
8. ***PARS INTERMEDIUS*** may be marked by small cystic follicles that are the remnants of Rathke's pouch. The follicles are often filled with a colloid. The function of the Pars Intermedius in humans is not well understood. They are positive for MSH and so are assumed to be corticotropes.
9. ***PARS TUBERALIS*** is an extension of the anterior lobe that wraps up and around the infundibulum. Functionally these cells stain positively for ACTH and LH and FSH and so would be classified as Gonadotropes and corticotropes.

- Examine the pars distalis (**Pituitary**, **Pituitary-1**, **and** **Pituitary-2**). The parenchyma of the anterior lobe (pars distalis) consists of cords of cells separated from one another by sparse connective tissue and by sinusoidal capillaries. In routinely stained sections, three cell types can be recognized. Identify acidophils, basophils and chromophobes. The acidophils are red and the basophils bluish while the chromophobes have an unstained, virtually translucent cytoplasm.

G. Posterior Pituitary Gland (Neurohypophysis)

1. Made of the infundibulum and the PARS NERVOSA.
2. Pars nervosa is not, technically, an endocrine gland. It is a location for the storage of neurosecretions.
3. Cell bodies of unmyelinated neurons are found in the supraoptic and paraventricular nuclei of the hypothalamus. The synaptic terminals of these neurons are in the pars nervosa and, when stimulated, release contents near fenestrated capillaries.
   a. The ends of the axons are typically large and filled with neurosecretory vesicles. These can sometimes be seen by light microscopy as dilations called Herring bodies, but they are difficult to identify in light microscopy.
   b. Neurosecretory vesicles of a single neuron in the Pars Nervosa contain either Oxytocin or Antidiuretic hormone.
4. Other cells in the pars nervosa are fibroblasts, mast cells and special glial cells called pituicytes. These glial cells resemble astrocytes and are situated near the fenestrated capillaries.
   - Examine the pars nervosa (**Pituitary-1, and** Pituitary-2). Identify the pituicytes. These are glia-like cells, with numerous cytoplasmic processes that surround axons from the hypothalamus and make up the largest cell population. Remember the secretory products from hypothalamic neurons are stored in the neurohypophysis in structures called Herring bodies.

H. Thyroid Gland

1. The thyroid is a bilobed gland in the anterior neck. The lobes are connected by a continuation of tissue called the isthmus. Each lobe possesses a thin capsule that invaginates and subdivides it into lobules.
2. The thyroid gland is derived, embryologically, from the floor of the pharynx. It begins as an invagination of tissue called the thyroglossal duct. The duct descends through the neck before splitting into lobes and coming to rest near the proximal trachea.
3. The endodermal cells of the duct arrange into plates of follicular cells. These will come to form larger colloid filled follicles.

4. Additional epithelial cells (ultimobranchial bodies) migrate into the thyroid glands from the 4th branchial pouches. These become the parafollicular cells.
5. The thyroid follicle is the functional unit. Follicular epithelial cells surround large pools of colloid substance.
   a. The follicular cells (principal cells) produce the T3 and T4 hormones and package them in colloid. They are also responsible for resorbing the colloid when thyroid hormones are needed to be released.
      i. Colloid is composed mostly of thyroglobulin, a large, iodinated glycoprotein that stores thyroid hormones in an inactive state.
      ii. Active hormones are liberated from the colloid by the follicular cells and transported via thyroid hormone transporter molecules (like MCT8 and MCT10) into the blood stream when stimulated.
      iii. In addition to the fenestrated capillaries the follicles are also surrounded by lymphatic vessels that may provide an alternate route for hormonal exit from the organ

   b. Parafollicular cells (C-cells) are along the periphery of the follicle and within the basal lamina. These cells secrete calcitonin and function in calcium metabolism. In H&E stains they have pale cytoplasm

- Examine **Thyroid Gland-1**, **Thyroid Gland-2** and **Parathyroid/Thyroid Gland** and note the thin connective tissue capsule whose projections invade the gland and partition it into poorly defined lobules. The interstitial tissue between the follicles will contain fenestrated capillaries.
- Examine the large thyroid follicles. See they are composed of a single layer of epithelial (follicle) cells enclosing a spheroidal space occupied by a homogeneous appearing colloid. These epithelial cells vary in height from low cuboidal or squamous to columnar depending on the active state of the follicle. Look for active follicles. In *active follicles*, the follicular epithelial cells are more columnar in morphology, and you may see endocytic vesicles in the apical regions. Occasionally, senescent epithelial cells slough off and become suspended within the colloid.
- Look for the parafollicular cells within the follicular epithelium or in the connective tissue spaces just adjacent to or between follicles. They tend to be larger than follicular epithelial cells and have pale cytoplasm.

I. Parathyroid

1. These are small nodules of epithelioid glandular tissue embedded in the posterior surface of the thyroid glands.
   a. If the thyroid is removed the parathyroid glands must be preserved if not preserved supplementation with calcium and Vit D is required

2. There are typically 4, in total and are derived from the 3rd and 4th pharyngeal pouches. Those from the 3rd pouch often get incorporated into the thymus, as that too is from the third pouch.

3. Two cell types exist in the parathyroid gland.
   a. Principle (Chief) cells are the most abundant of the parenchymal cells in the parathyroid glands and they make parathyroid hormone (PTH). They are small polygonal cells with central nuclei. They tend to be chronically stimulated by changes in blood calcium levels.
      i. Because of the high need for calcium metabolism in fetal life the chief cells of the parathyroid gland differentiate and are very active during fetal life.

   b. Oxyphil cells are larger and more acidophilic, they accumulate throughout life and are more abundant in tissue from older individuals. They have been shown to secrete PTH like principal cells but additional functions or the reason for their altered appearance is not known.
      - Examine the slides **Parathyroid Gland-1** and **Parathyroid/Thyroid gland**, the parenchymal cells form irregularly shaped, anastomosing cords or groups. Two cell types can be identified: the principal or chief cell and the oxyphil cell. The chief cells are polyhedral and slightly acidophilic with a round vesicular nucleus. They produce parathyroid hormone. You may be able to further distinguish between dark and light chief cells with the darker cells containing more membrane bound granules and representing a more active secretory stage.
      - **Oxyphil** cells appear at the onset of puberty and are fewer in number. They are larger than chief cells, with smaller and darker nuclei. Their cytoplasm is acidophilic. The distinction between the two cell types is particularly good in **Parathyroid Gland-2**.

J. Adrenal Gland

1. These are paired encapsulated organs located within the perirenal fascia in the retroperitoneal space.
2. Each gland possesses a cortex, derived from embryonic mesoderm and a medulla, derived from embryonic neural crest cells which have migrated into the gland.
3. Each has numerous arteries that supply the gland and a single vein that drains. Within the organs the microvasculature is unique, too.
    a. Cortical arterioles branch from the capsular arteries to supply the cortex via fenestrated sinusoidal capillaries. This blood trickles through the cortical tissue and into the medulla before draining to the central medullary vein.
    b. Medullary arterioles branch from the capsular vessels and move directly to the medullary cells. Hence the medulla gets a dual blood supply, one fresh of the capsular artery and one conditioned and containing the secretions from the cortex.
4. Lymphatics are found in the capsule and with larger vessels and are needed to distribute the chromogranin A from the chromaffin cells.

K. Adrenal Medulla

1. The medulla is populated by chromaffin cells.
2. These are modified neurons and neural crest cell derived. And they act like a giant sympathetic ganglion.
    a. They lack axons, likely due to the actions of the glucocorticoids from the cortex. They do not synapse anywhere, rather they secrete their neurotransmitters into the fenestrated capillaries.
3. Pre-synaptic sympathetic neurons enter the medulla with the vasculature.
4. Chromaffin cells are highly secretory and makes/secrete either epinephrine or norepinephrine.
    a. Fight or flight response causes numerous metabolic changes.
        i. Glycogenolysis
        ii. Fatty acid mobilization, increase BP, select vasodilation, etc.

L. Adrenal cortex

1. The cortex of the adrenal gland is separated into three functional zones. From superficial to deep, they are:
    a. Zona glomerulus – thin outer regions
    b. Zona fasciculata – largest area
    c. Zona reticularis – inner near medulla comprising about 5%

2. Zona Glomerulus cells are tightly packed in clusters, and they are continuous with the cords of the fasciculata region but have different physiology.
    a. Very metabolically active with lots of smooth endoplasmic reticulum (ER) and Golgi complexes
    b. They have abundant fenestrated capillaries.
    c. Cells of the Zona glomerulus make mineralocorticoids, like aldosterone for sodium and potassium and water balance.
    d. Activity of these cells is regulated by the RAAS system.

3. Zona fasciculata cells are arranged in long cords that span the space between the outer and inner cortical zones.
    a. Histologically the cells are acidophilic with lots of smooth ER and the cytoplasm has lots of lipid droplets.
    b. They secrete glucocorticoid steroids, primarily, and a small amount of the gonadocorticoids.
        i. The major glucocorticoid is cortisol.
        ii. The glucocorticoids have far reached and diverse action on metabolism at the systemic and cellular levels.

    c. Activity of the cells of the zona fasciculata is regulated by feedback loops, specifically CRH-ACTH
        i. ACTH stimulates the fasciculata cells, increase blood flow to the area and is linked to hypertrophy.
        ii. Circulating levels of cortisol modulated hypothalamic release of CRH which controls ACTH secretion from the pituitary.

4. Zona reticularis cells are smaller and more densely packed. They are more darkly stained with less cytoplasm and fewer lipid droplets, though they have a lot of smooth ER.
   a. They secrete, mostly, the gonadocorticoids (adrenal androgens)
      i. Dehydroepiandrosterone (DHEA)
      ii. Dehydroepiandrosterone sulfate (DHEAS)
      iii. Androstenedione

   b. In patients assigned male at birth, these adrenal androgens have little function as testosterone is a more powerful androgen.
   c. In patients assigned female at birth these adrenal androgens promote different aspects of maturation during puberty.
      i. DHEA can be, ultimately, converted to testosterone and estrogens in peripheral tissues.
   d. Activity of the cell sin the zona reticularis is under the control of the CRH_ACTH feedback loops.

- Examine Adrenal Gland-1**, and** Adrenal Gland-2, the adrenal gland (cortex and medulla) is two endocrine tissues enclosed within a common capsule.
- Examine the **adrenal medulla** on these slides for the irregularly shaped **chromaffin** cells containing membrane bound granules (catecholamines). Yellow lipofuscin pigment granules may be seen. The chromaffin cells are arranged in groups of short cords separated by capillaries and venules.
- Identify on these slides the three concentric zones of the adrenal cortex. (1) The zona glomerulosa is the outer zone (beneath the capsule). It is composed of spherical groups of columnar cells. This zone blends into the subjacent zona fasciculate. (2) The zona fasciculata is the central zone, composed of radially arranged cords or columns of polyhedral cells separated by long sinusoidal blood vessels. Numerous vacuoles are often present in the cells of this zone representing lipid droplets dissolved away during histological preparation. These vacuoles occur with somewhat lesser frequency in the other two zones of the cortex. (3) The zona reticularis is the innermost zone of the cortex consisting of a compact anastomosing network of cellular cords with evidence of lipid droplets. The nuclei in all three zones are vesicular and spherical.

- Return to the capsular region of these sections through the adrenal glands and examine the microvascular networks. Identify the **capsular arteries** forming a sinusoid plexus within the capsule itself. From these vessels arise two vascular systems. The **cortical sinusoids** supply the cortex and can be seen separating cords of the zona fasciculata. The **medullary arteries** penetrate the cortex via connective tissue trabeculae to supply the large medullary sinusoids.

M. Embryology/Fetal Adrenal Gland

1. The adrenal gland is very important in utero.
2. The cortex takes up more area in the fetal gland and is referred to as the fetal zone.
   a. The Permanent cortex will arise from an area along the periphery of the gland near the capsule.
   b. The fetal gland lacks a large defined medulla as the chromaffin cells have not fully amassed.
3. Activity of the cells in the fetal zone is highly regulated by the pituitary CRH-ACTH axis.
4. The fetal adrenal gland works in conjunction with the placenta for steroid synthesis as each express distinct enzymes needed in the synthesis pathways.
   a. This Fetal-Placental unit transports precursor molecules back and forth to manufacture glucocorticoids, aldosterone, androgens, and estrogens.
5. The fetal cortex rapidly involutes early in postnatal life
   a. Failure of involution leads to CAH congenital adrenal hyperplasia.
      i. 21-hydroxyolase deficiency is the most common genetic cause of CAH

Answers to Study Questions

These answers may not be all encompassing; rather they may be examples of the types of thought processes needed for application of the material.

1. Compare and contrast the embryonic origins of the pituitary lobes with respect to their histological appearance.

The anterior pituitary gland is derived from the pharyngeal **ectoderm**. This impacts the histology such that the cells are epithelioid in nature. They look much like other epithelial-derived glands, like the salivary glands. Like other glands the cells are arranged in clusters and have multiple physiologies, which is reflected in their staining patterns. The posterior lobe is derived from neural tissue and resembles the brain when stained with H&E. It has a central cavity, continuous with the third ventricle, and similarly lined with an epithelium. There are glial cells, pituicytes, that resemble astrocytes in form and function. Also, like neural tissues there are numerous axons and axon terminals, Herring bodies, engorged with neurohormones ready for release. What is not seen in the posterior lobe of the pituitary gland is neuronal cell bodies. The cell bodies of pituitary neurons are in the hypothalamic tissues.

2. One of your patients has mutation that drives the expression of 17-α-hydroxylase expression in the Zona glomerulus. What might be some metabolic consequences?

The zona glomerulus does not typically express this enzyme. In other regions of the adrenal cortex this enzyme Is highly expressed and is needed for the synthesis of cortisol and testosterones and estrogens. It is not needed for the biosynthesis of aldosterone. If the zona glomerulus were to express this enzyme, then the cells would be able to generate androgens and cortisol. This may lead to increased cortisol secretion and mimic disorders like Cushing's disease. It may drive masculinization or precocious puberty Because cholesterol, in the zona glomerulus cells, is now being diverted to the steroid pathways aldosterone levels may drop reducing sodium resorption in the kidney and reducing potassium secretion, as well, mimicking chronic inhibition of the RAAS system. Histologically the tissues may look different as well. The cells in the outer glomerulus zone may begin to resemble the zona fasciculata or reticularis depending on which metabolic pathway predominates, cortisol v androgen.